Ex/EE/PC/B/T/223/2025

B.E. ELECTRICAL ENGINEERING SECOND YEAR SECOND SEMESTER EXAM 2025

Time: Three hours **Full Marks: 100**

POWER SUPPLY SYSTEMS

(50 marks for each part)

Use separate answer script for each part.

PART I

Solutions

May 16, 2026

Question 1(a)

Problem Statement: What do you understand by pulverized fuel firing? Discuss about its advantages and disadvantages[cite: 9].

Thought Process

Goal: Define pulverized fuel firing and evaluate its pros and cons.
Methodology: 1. Define the process: crushing coal into a fine powder to increase surface area.
2. Explain the firing process: injecting the powder with primary air into the furnace.
3. List advantages: improved thermal efficiency, faster response times, and ability to burn low-grade coal.
4. List disadvantages: high capital cost for milling equipment, increased fly ash generation, and explosion risks.

Answer:
Understanding Pulverized Fuel Firing:
Pulverized fuel firing is a method of burning coal in thermal power plants where the raw coal is crushed and ground into a fine powder (dust) before being injected into the combustion chamber. The pulverization significantly increases the surface area of the coal particles. This fine coal dust is mixed with pre-heated primary air and blown into the furnace, where it burns rapidly and almost completely in suspension, resembling the combustion of a gas.

Advantages:

- **Higher Thermal Efficiency:** The increased surface area allows for complete and rapid combustion, minimizing carbon loss in ash.
- **Rapid Response to Load Changes:** Because the fuel burns almost instantaneously, the heat release rate can be rapidly adjusted to meet fluctuating load demands.
- **Utilization of Low-Grade Coal:** It allows for the efficient combustion of low-grade coal with high ash content that would be difficult to burn on solid grates.
- **Lower Excess Air Requirement:** Intimate mixing of air and fuel requires less excess air for complete combustion, reducing heat loss in exhaust gases.

Disadvantages:

- **High Capital and Operating Cost:** The installation of ball mills, crushers, and handling equipment requires significant investment, and the pulverizing process consumes a considerable amount of electrical power.
- **Fly Ash Problem:** A large portion of the ash remains in suspension and is carried away by flue gases, requiring highly efficient and expensive gas cleaning equipment (like ESPs).
- **Explosion Risk:** Fine coal dust mixed with air is highly explosive, requiring strict safety measures and controlled environments during storage and transport.

Question 1(b)

Problem Statement: Name one accessory in steam power plant that can reduce the problem of 'fly ash'[cite: 10]. Discuss working principle of the equipment[cite: 11].

Thought Process

Goal: Identify an ash-reduction accessory and explain how it works.
Methodology: 1. Identify the equipment: The Electrostatic Precipitator (ESP). 2. Analyze the working principle: Explain the generation of a high-voltage corona, ionization of gases, charging of ash particles, and their subsequent collection on grounded plates.

Answer:
Equipment Name:
An essential accessory used in modern steam power plants to mitigate the fly ash problem is the **Electrostatic Precipitator (ESP)**.

Working Principle of the Electrostatic Precipitator:
The Electrostatic Precipitator utilizes the principles of electrostatics to remove fine particulate matter (fly ash) from the flowing exhaust flue gas.

1. **Ionization:** The ESP consists of discharge electrodes (typically thin wires) and collection electrodes (grounded flat plates). A highly negative direct current (DC) voltage (usually between 20 kV to 100 kV) is applied to the discharge wires. This creates a strong electric field, causing the gas immediately surrounding the wires to ionize, generating a corona discharge that releases a heavy stream of electrons.
2. **Particle Charging:** As the dust-laden flue gas flows through the electric field, the free electrons and ionized gas molecules attach themselves to the fly ash particles, giving the ash a strong negative charge.
3. **Collection:** The negatively charged fly ash particles are driven by the electric field towards the grounded (positively charged relative to the wires) collection plates. The particles accumulate on these plates.

4. **Rapping (Removal):** Periodically, mechanical hammers or "rappers" strike the collection plates. The physical shock dislodges the agglomerated ash, which falls due to gravity into collection hoppers located at the bottom of the unit, allowing clean gas to pass to the chimney.

Question 2

Problem Statement: What are the desirable features of a peak load plant? [cite: 12] Discuss why steam power plant is preferred to operated as base load plant but gas turbine plant is preferred to operate as peak load plant[cite: 13].

Thought Process

Goal: Detail peak load plant characteristics and justify operational roles for steam vs. gas plants.
Methodology: 1. Identify peak load features: Quick start/stop, low capital cost, fast synchronization, and ability to handle variable loads.
2. Compare steam vs. gas: Steam has high thermal inertia (takes hours to start) but low running cost (coal), ideal for continuous base load. Gas turbines start within minutes and have lower capital costs, but use expensive fuel, making them economically and technically suited only for short, peak durations.

Answer:
Desirable Features of a Peak Load Plant:
A peak load plant operates only during the hours of maximum power demand. Therefore, it must possess the following features:

- **Quick Start-up and Synchronization:** It must be able to start from cold conditions and synchronize with the grid within minutes to respond to sudden load spikes.
- **Low Capital Cost:** Since it operates for only a few hours a day (low load factor), fixed capital costs must be minimized to keep the overall cost of generated energy reasonable.
- **Fast Responsiveness:** It must be highly responsive to load fluctuations and capable of rapid ramping up or down.
- **Standby Readiness:** It should incur minimal to no standby losses when not generating power.

Why Steam Plants are Base Load and Gas Turbines are Peak Load:

- **Steam Power Plant (Base Load):** Steam plants have immense thermal inertia. Heating hundreds of tons of water in massive boilers to generate high-pressure superheated steam takes hours. Consequently, a steam plant cannot respond quickly to sudden demand spikes. Furthermore, they are highly capital-intensive but utilize relatively cheap fuel (coal). Economically, to recover the massive initial investment, the plant must operate continuously at high load factors. Thus, both technically (slow start) and economically (high capital/low running cost), steam plants are heavily preferred for base load.

- **Gas Turbine Plant (Peak Load):** Gas turbines operate on the Brayton cycle and do not require massive boilers or steam generation. The compressor and turbine can be spun up by a starting motor, ignite the fuel, and reach full capacity within 10 to 15 minutes. While their capital cost is significantly lower than steam plants, their running cost is very high due to the expensive fuel (natural gas or high-grade liquid fuels). Running them continuously would be economically disastrous. Therefore, their rapid starting capability and low capital costs make them perfectly suited to cover short-duration peak loads.

Question 3

Problem Statement: With a suitable schematic diagram, discuss working principle of a nuclear power plant having pressurized water reactor[cite: 16].

> Thought Process
>
> **Goal:** Illustrate and explain a Pressurized Water Reactor (PWR) nuclear plant.
> **Methodology:** 1. Draw a block diagram showing the Primary Loop (Reactor core, Pressurizer, Coolant Pump) and Secondary Loop (Steam Generator, Turbine, Condenser).
> 2. Explain the two-circuit architecture: The primary circuit uses water under extreme pressure (to prevent boiling) as a coolant and moderator.
> 3. Explain heat exchange: Primary water transfers heat to the secondary circuit in the steam generator, boiling the secondary water to drive the turbine.

Answer:
Schematic Diagram of a PWR Plant:

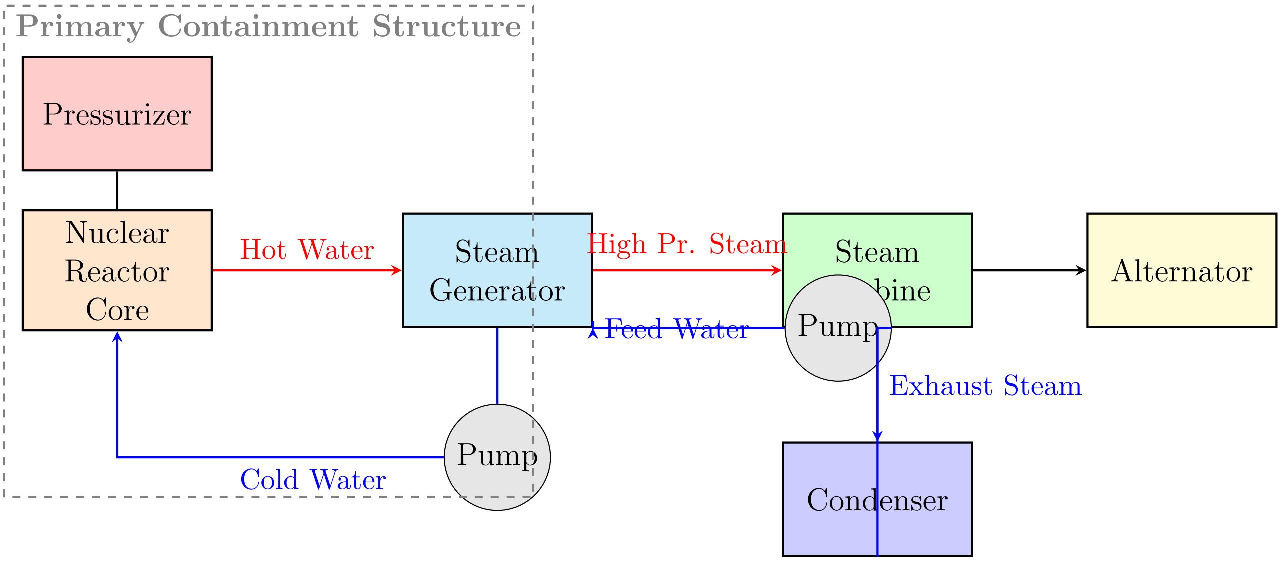


Figure 1: Schematic of a Pressurized Water Reactor (PWR) Nuclear Power Plant

Working Principle:
The Pressurized Water Reactor (PWR) is characterized by its distinct two-loop system (primary and secondary), ensuring that radioactive materials remain entirely confined within the containment structure.

1. **Primary Circuit:** Ordinary (light) water acts as both the coolant and the moderator within the reactor core. Nuclear fission generates immense heat. The primary coolant pump circulates water through the reactor core to absorb this heat. A **Pressurizer** maintains the primary circuit at a very high pressure (around 150 atmospheres). This extreme pressure prevents the water from boiling, allowing it to remain liquid even at temperatures exceeding 300°C.

2. **Heat Exchange:** The super-heated, highly pressurized primary water flows through U-tubes inside the **Steam Generator**.

3. **Secondary Circuit:** Feedwater in the secondary circuit flows over the outside of these U-tubes. The heat transfers from the radioactive primary water to the non-radioactive secondary water. Because the secondary circuit is at a much lower pressure, the absorbed heat causes the secondary water to boil and vaporize into steam.

4. **Power Generation:** The high-pressure steam generated is piped to the steam turbine, causing it to spin. The turbine drives an alternator to produce electricity.

5. **Condensation:** The spent steam exiting the turbine is condensed back into liquid water inside the condenser via external cooling water, and pumped back to the steam generator to repeat the cycle.

Question 4

Problem Statement: Explain the significance of the load curve[cite: 17]. The variation of the load over the 24 hours of a day for a certain area is as given below[cite: 19]. Draw the load curve and load-duration curve. Calculate average load, load factor and demand factor[cite: 20, 21]. (Table data [cite: 24]).

Thought Process

Goal: Define load curve significance, plot graphs from data, and calculate metrics.
Methodology: 1. Significance: Helps in sizing generation units, calculating energy generated, predicting peak demand, and scheduling maintenance.
2. Data Mapping (0 to 24 hrs starting at midnight): 0-3: 40MW, 3-6: 20MW, 6-9: 30MW, 9-12: 50MW, 12-15: 70MW, 15-18: 70MW, 18-21: 120MW, 21-24: 80MW.
3. Load Duration Curve: Sort loads descending: 120MW(3h), 80(3h), 70(6h), 50(3h), 40(3h), 30(3h), 20(3h). Plot cumulative hours.
4. Calculations: Total Energy (Area) = $\sum(P \times t) = 1440$ MWh. Average Load = Total Energy / 24. Load Factor = Average Load / Max Demand. Demand Factor = Max Demand / Connected Load. (Note: Since Connected Load is not provided, state the assumption that Connected Load $\geq$ Max Demand).

Answer:

(a) Significance of the Load Curve:

The load curve is a graphical representation showing the variation of electrical load on a power station with respect to time. Its significance includes:

- **Area under the curve:** Represents the total electrical energy generated or consumed in the given period.
- **Plant Operation:** It identifies the maximum demand, which dictates the installed capacity of the plant.
- **Unit Selection:** By analyzing base and peak loads, engineers can decide the size and number of generating units required to operate efficiently.
- **Cost Optimization:** It is utilized to calculate load factors, directly impacting the economics and unit cost of generation.

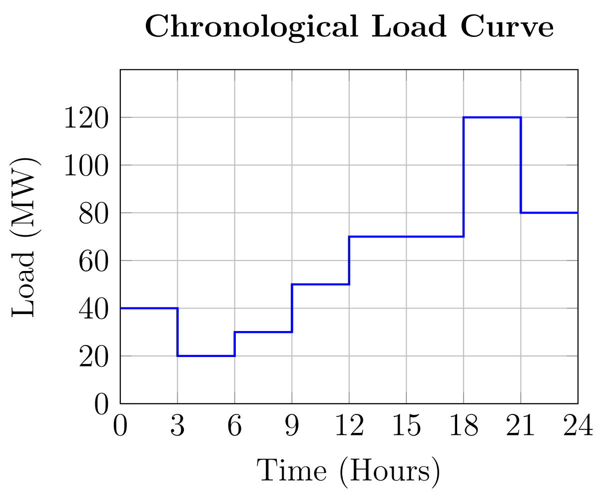
(b) Curves and Calculations:

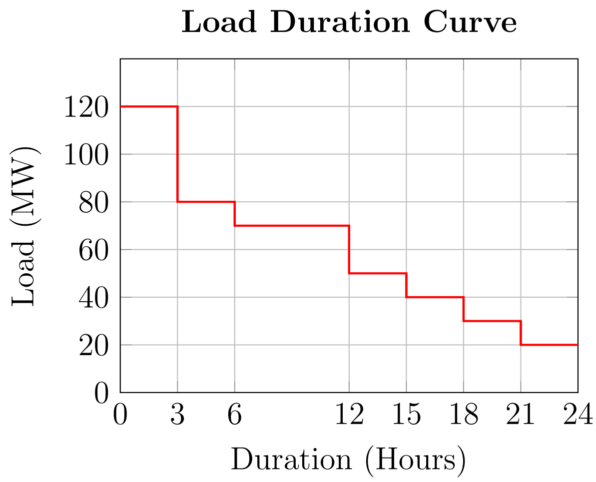
First, let us align the 24-hour tabular data starting from Midnight (00:00 Hours) to form the coordinates for our curves:

- 00:00 to 03:00 (12AM-3AM): 40 MW
- 03:00 to 06:00 (3AM-6AM): 20 MW
- 06:00 to 09:00 (6AM-9AM): 30 MW
- 09:00 to 12:00 (9AM-12noon): 50 MW
- 12:00 to 15:00 (12noon-3PM): 70 MW
- 15:00 to 18:00 (3PM-6PM): 70 MW

- 18:00 to 21:00 (6PM-9PM): 120 MW
- 21:00 to 24:00 (9PM-12AM): 80 MW

Load Curve & Load Duration Curve:





Calculations:

1. **Total Energy Generated (Area under curve):**

$$\begin{aligned}
\text{Energy} &= \sum(\text{Load} \times \text{Time}) \\
&= (40 \times 3) + (20 \times 3) + (30 \times 3) + (50 \times 3) + (70 \times 6) + (120 \times 3) + (80 \times 3) \\
&= 120 + 60 + 90 + 150 + 420 + 360 + 240 \\
&= 1440 \text{ MWh}
\end{aligned}$$

2. **Average Load:**

$$\text{Average Load} = \frac{\text{Total Energy}}{\text{Total Hours}} = \frac{1440 \text{ MWh}}{24 \text{ hours}} = 60 \text{ MW}$$

3. **Load Factor:**

$$\text{Load Factor} = \frac{\text{Average Load}}{\text{Maximum Demand}} = \frac{60 \text{ MW}}{120 \text{ MW}} = 0.5 \text{ or } 50\%$$

4. **Demand Factor:**

$$\text{Demand Factor} = \frac{\text{Maximum Demand}}{\text{Total Connected Load}}$$

Note: The problem does not provide the "Connected Load". If we assume the connected load is equal to the maximum demand (a common assumption when data is missing in purely theoretical exercises), the Demand Factor would be $120/120 = 1.0$ (or 100%). However, strictly speaking, it cannot be calculated without the total connected load data.

Question 5

Problem Statement: With the help of a schematic diagram discuss the working principle of a closed cycle gas turbine plant[cite: 25].

Thought Process

Goal: Explain a closed-cycle gas turbine layout and principle.
Methodology: 1. Identify components: Compressor, External Heater (Heat Exchanger), Turbine, Pre-cooler.
2. Draw a block diagram forming a closed loop.
3. Explain the Brayton cycle dynamics: Fluid (often Helium or air) is compressed, heated externally (no mixing with combustion products), expanded in the turbine to do work, and cooled before re-entering the compressor. This isolates the working fluid from fuel impurities.

Answer:
Schematic Diagram:

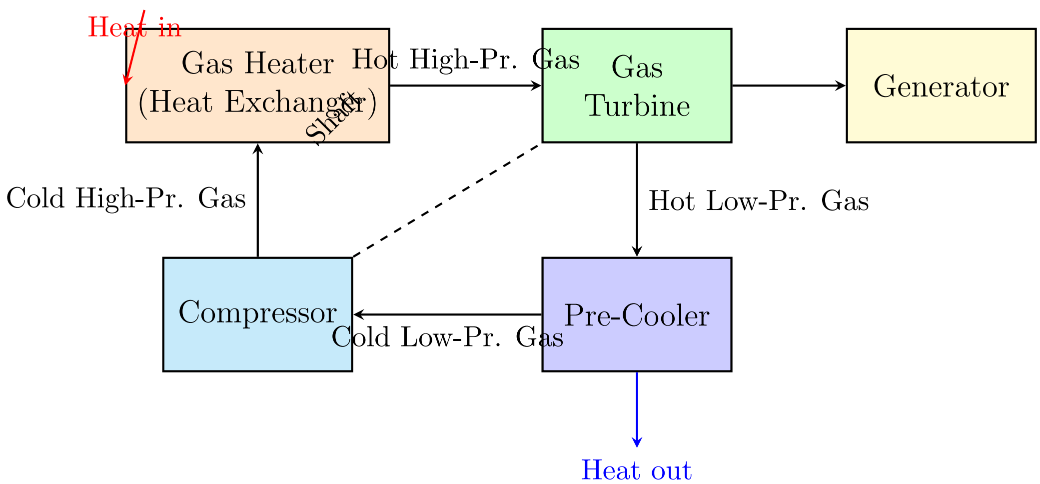


Figure 2: Schematic of a Closed Cycle Gas Turbine Plant

Working Principle:
A closed-cycle gas turbine operates continuously on the thermodynamic Joule/Brayton cycle using a constant mass of working fluid (usually air, helium, or argon) circulating in a completely sealed loop.

1. **Compression:** The working fluid enters the rotary compressor at a low pressure and low temperature. The compressor performs work on the gas, significantly increasing its pressure.

2. **External Heating:** The high-pressure cold gas enters a heat exchanger (Gas Heater). Unlike an open cycle where fuel is burned directly in the air, here heat is transferred to the working fluid from an external source (such as a coal furnace or a nuclear reactor). Because combustion products never mix with the working fluid, turbine blade fouling is eliminated.

3. **Expansion:** The high-pressure, high-temperature gas flows into the gas turbine and expands. The expansion exerts force on the turbine blades, generating mechanical energy to drive both the alternator (for electricity) and the compressor (via a common shaft).

4. **Cooling:** The hot, low-pressure gas exiting the turbine enters a pre-cooler (another heat exchanger), where circulating cooling water absorbs the waste heat. The gas is returned to its initial low temperature and directed back into the compressor to complete the cycle.

Question 6

Problem Statement: Write a comparative study among Pelton wheel, Francis turbine and Kaplan turbine[cite: 26]. What are the factors to be considered in site selection of hydroelectric power plant? [cite: 27]

Thought Process

Goal: Compare the three main hydro turbines and list hydro plant site selection criteria.
Methodology: 1. Turbine Comparison: Create a table comparing Head (High vs Medium vs Low), Flow direction (Tangential, Mixed, Axial), Type (Impulse vs Reaction), and Specific Speed.
2. Site Selection Factors: Water availability, head of water, geological strength, reservoir storage capacity, environmental impact, and transmission distance.

Answer:
(a) Comparative Study of Hydraulic Turbines:

Parameter	Pelton Wheel	Francis Turbine	Kaplan Turbine
Type of Turbine	Impulse Turbine	Reaction Turbine	Reaction Turbine
Water Head	High Head (> 250m)	Medium Head (30m to 250m)	Low Head (< 30m)
Discharge Quantity	Low discharge	Medium discharge	High discharge
Flow Direction	Tangential flow	Mixed flow (Radial in, Axial out)	Axial flow
Specific Speed	Low (10 to 50)	Medium (50 to 250)	High (250 to 850)
Runner Blades	Fixed buckets	Fixed profile blades	Adjustable pitched blades
Draft Tube	Not required	Required to recover pressure	Required (highly critical)

(b) Factors Considered in Site Selection of a Hydroelectric Plant:

- **Availability of Water:** The fundamental requirement. Extensive study of historical hydrographs and mass curves is necessary to guarantee sufficient continuous water discharge throughout the year.
- **Water Head:** A large water head minimizes the size of the required reservoir and turbines for a given power output. The site geometry should naturally accommodate a significant drop.
- **Geological and Foundation Conditions:** The rock structure at the site must be strong and stable enough to support the immense weight of the dam and withstand water thrust and seismic forces without seepage.
- **Topography for Reservoir:** A narrow canyon or gorge opening into a wide valley upstream is ideal, minimizing the dam size while maximizing water storage capacity.
- **Distance from Load Center:** Because suitable natural sites are often remote, the cost of erecting long, high-voltage transmission lines to populated load centers must be economically viable.
- **Accessibility and Environmental Impact:** The site must be accessible for heavy construction machinery. Additionally, the ecological and social impact of flooding land for the reservoir must be assessed.

Question 7

Problem Statement: Discuss the functions of (i) moderator in nuclear power plant, (ii) compressor in gas turbine power plant, (iii) surge tank in hydroelectric power plant,

(iv) superheater of steam power station, (v) starting motor in gas turbine plant[cite: 28, 29].

Thought Process

Goal: Define the precise mechanical/nuclear function of five specific components. **Methodology:** 1. Moderator: Slows neutrons to sustain fission. 2. Compressor: Supplies high-pressure air for the Brayton cycle. 3. Surge Tank: Relieves transient pressure (water hammer) in penstocks. 4. Superheater: Removes moisture, increases efficiency, protects turbine. 5. Starting motor: Initiates Brayton cycle until combustion becomes self-sustaining.

Answer:

1. **Moderator in a Nuclear Power Plant:**
During nuclear fission, the neutrons released are "fast neutrons" with very high kinetic energy. The probability of these fast neutrons causing further fission in Uranium-235 is very low. The moderator (like graphite or heavy water) acts as a physical medium to collide with these neutrons, slowing them down to "thermal" speeds without absorbing them, thereby sustaining a controlled chain reaction.

2. **Compressor in a Gas Turbine Power Plant:**
The compressor continuously draws in ambient air and compresses it to a very high pressure before delivering it to the combustion chamber. This high-pressure environment is required for the Brayton thermodynamic cycle, ensuring that when the fuel is injected and ignited, the rapid expansion generates maximum thrust/work through the turbine.

3. **Surge Tank in a Hydroelectric Power Plant:**
The surge tank is a vertical pipe/reservoir installed between the dam and the turbine. Its function is to protect the penstock from bursting due to "water hammer." If the turbine suddenly rejects load, the governor closes the water gates; the massive momentum of the rushing water creates high back-pressure, which is absorbed by the water surging up into the tank. Conversely, it provides immediate extra water when the turbine suddenly requires more load.

4. **Superheater of a Steam Power Station:**
The superheater takes the saturated (wet) steam produced in the boiler drum and heats it further above its saturation temperature using flue gases. This serves two functions: it significantly increases the thermal efficiency of the Rankine cycle, and it ensures the steam is completely dry. Dry steam is critical because moisture droplets at high velocities would severely erode the turbine blades.

5. **Starting Motor in a Gas Turbine Plant:**
Unlike a steam turbine, a gas turbine is not self-starting. The compressor must be rotating at a minimum specific speed to supply enough compressed air to ignite the fuel and sustain the continuous combustion cycle. The starting motor (often electric or a small diesel engine) physically drives the main shaft to spin up the compressor until the turbine reaches its self-sustaining speed, after which the motor is disengaged.